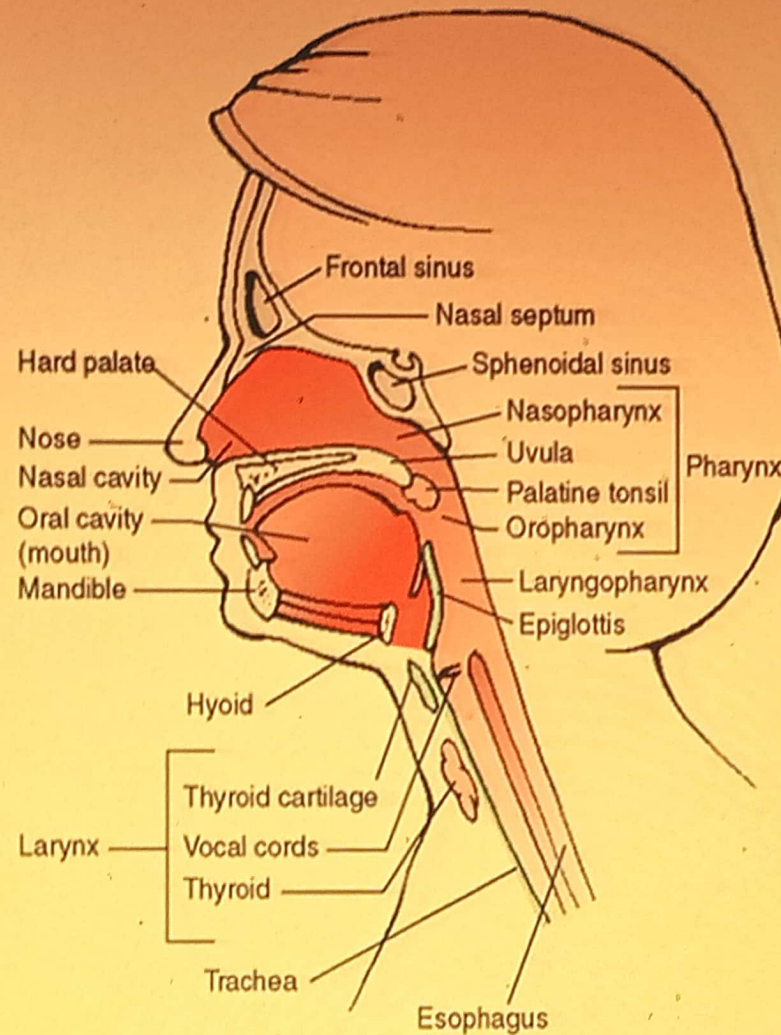
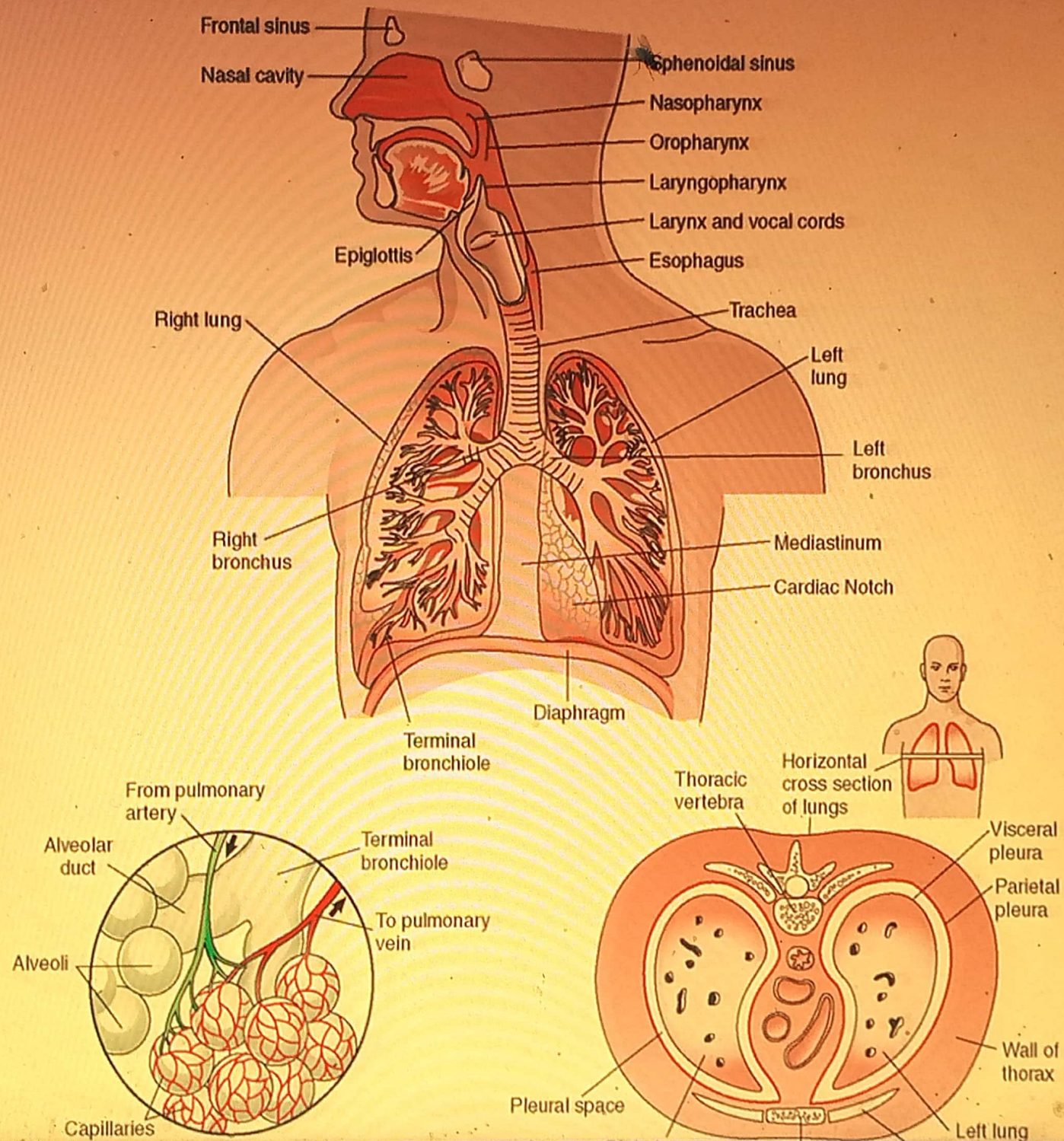
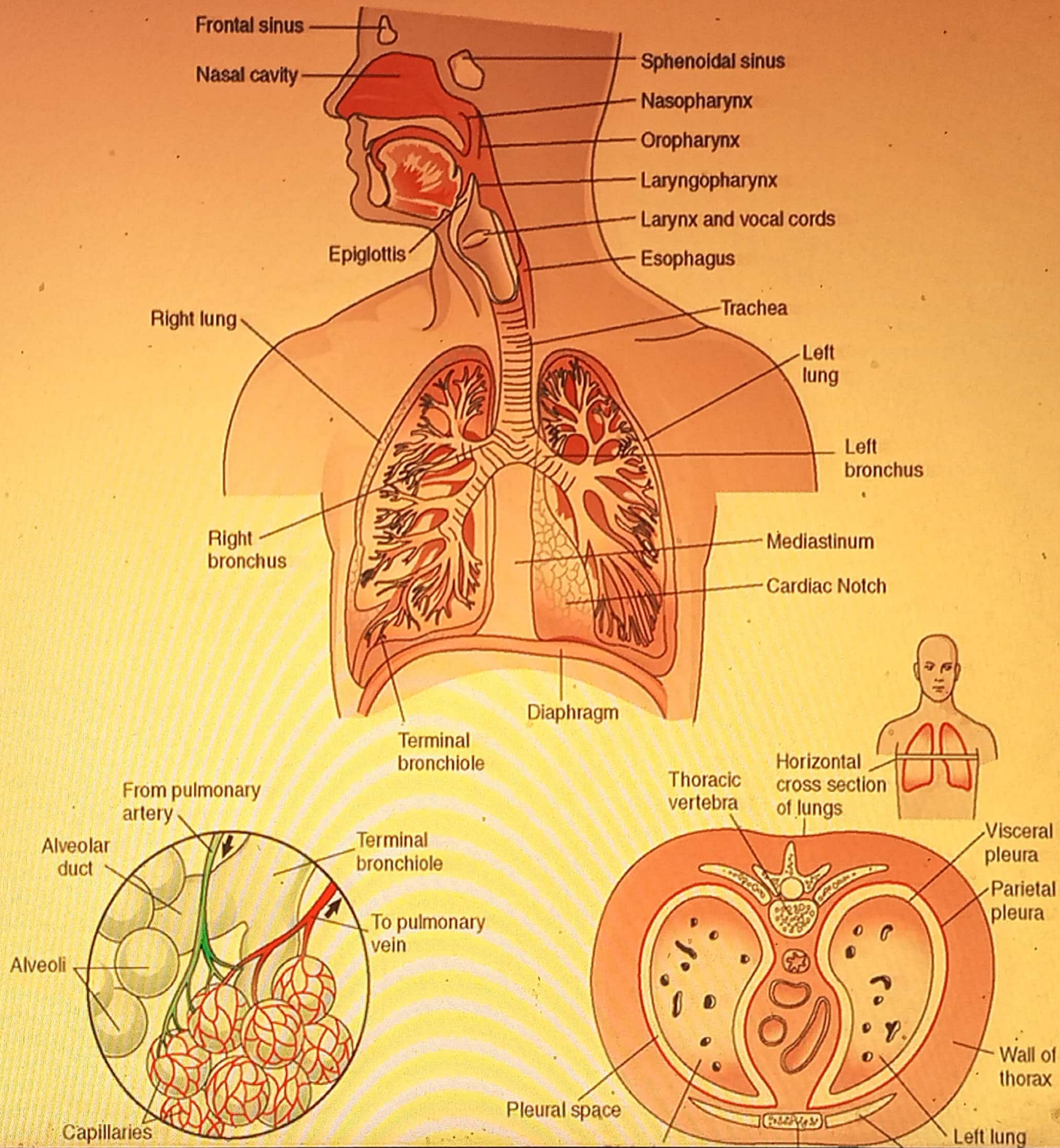


Respiratory System







Respiratory System

- The respiratory system is divided into:

1- Conducting portion, consisting of the **nasal cavity, nasopharynx, larynx, trachea, bronchi, bronchioles** and **terminal bronchioles**.

1, provide a series of air passages for moving the air to and from the gas exchange area in the lung.

2, Condition the inspired air before it enters the lung.

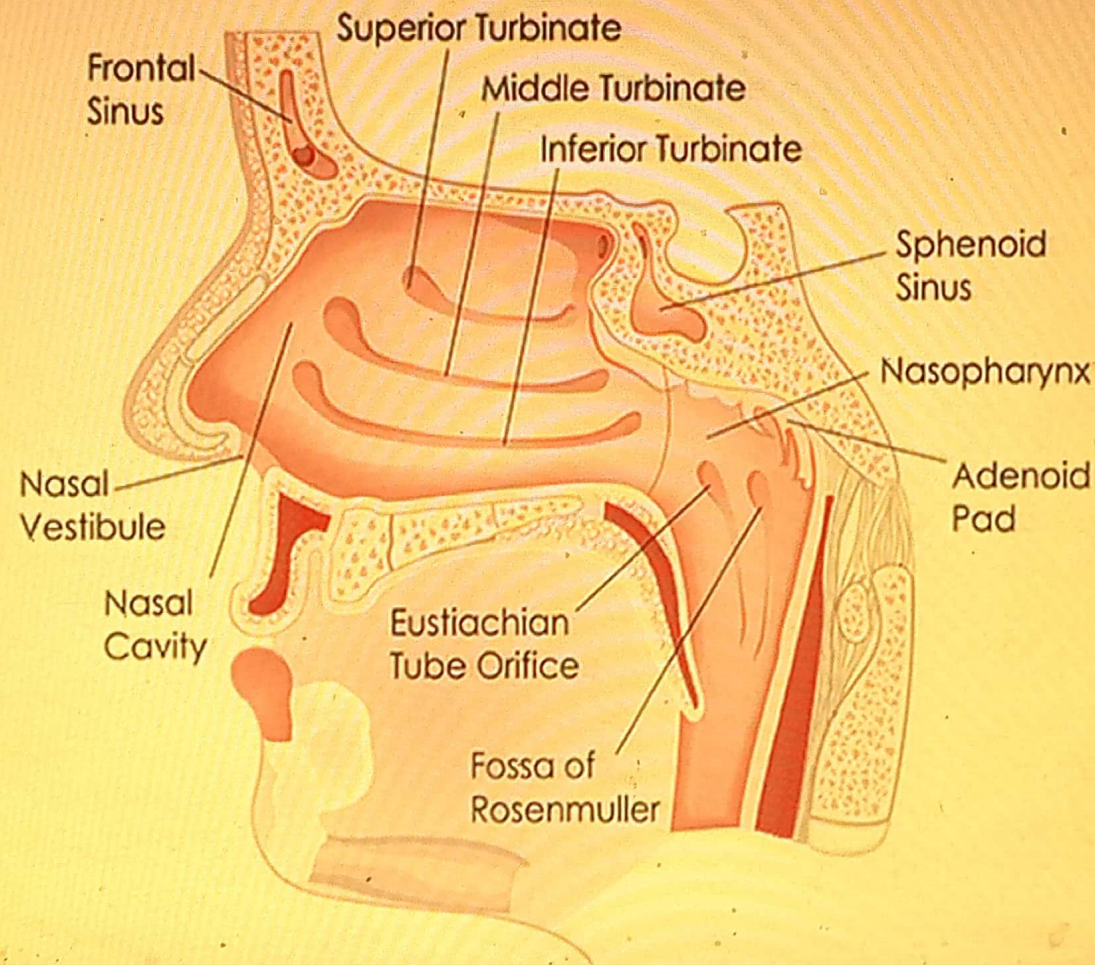
2- Respiratory portion, consisting of **respiratory bronchioles, alveolar ducts** and **alveoli**.

Nasal cavity

1- Nasal Vestibule

Rostrally, is lined by a pigmented thick keratinized stratified squamous epithelium. **At midvestibule**, the epithelium is thinner and non keratinized.

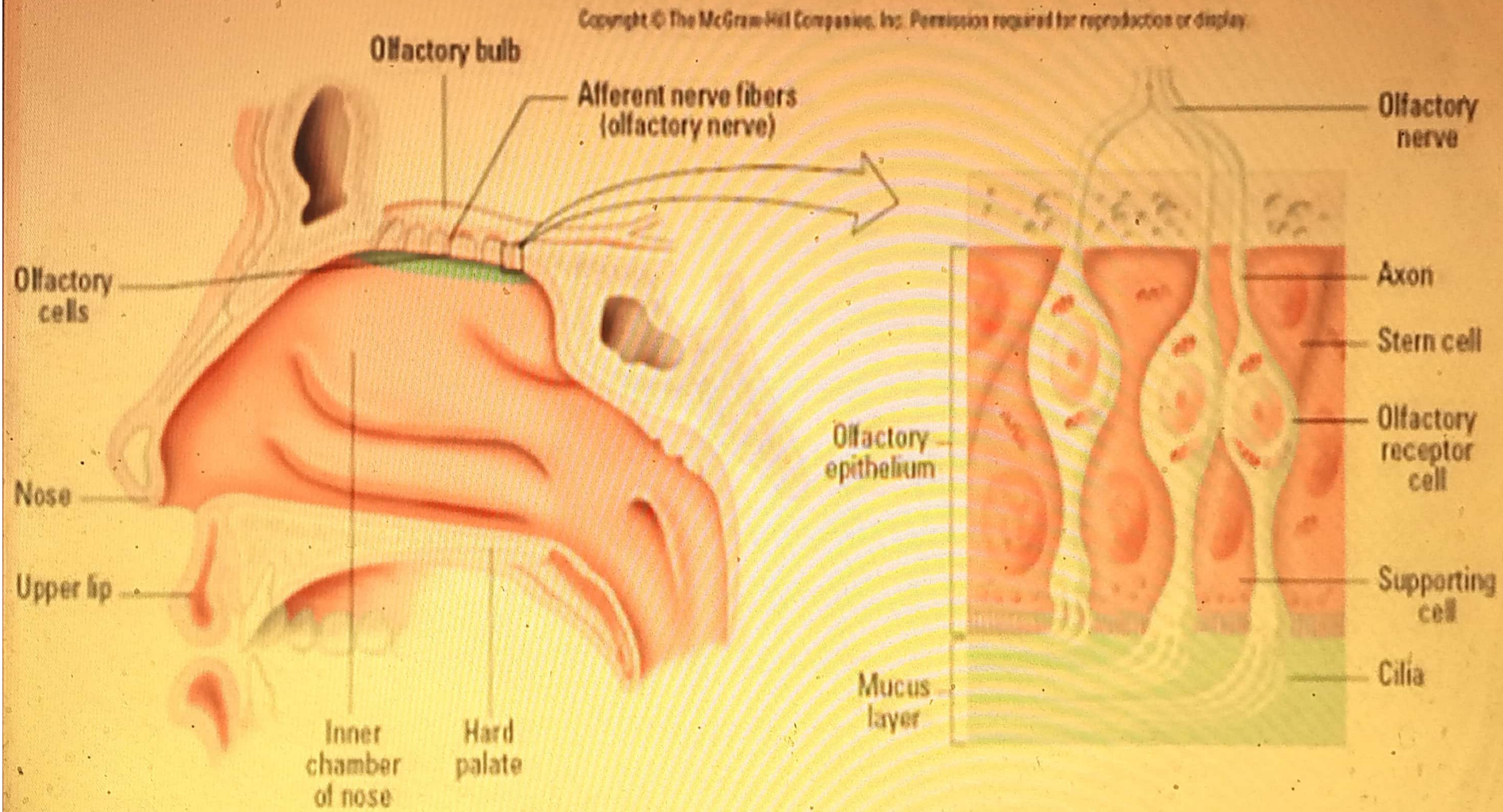
2- Nasal Cavity Proper. The lumen of the nasal cavity proper is subdivided by lamellae of the ventral, dorsal and ethmoid conchae.



Olfactory mucosa covers the caudodorsal portion of the nasal cavity.

- The epithelium of this mucosa is very thick and has pigment granules making it appears yellow to green-brown.

- It is lined by pseudostratified columnar epithelium consisting of basal, sensory and supporting cells.



Basal cells

- They are small, irregular or cone-shaped forming a single layer at the base of the epithelium.

Sensory olfactory cells (receptor cells)

- They are bipolar neurons with dendrites reaching the nasal cavity lumen and form club-shaped apices (dendritic bulbs) protrude into the lumen. 10-30 non-motile, very long cilia arise from each bulb and are considered to be receptors that respond to odoriferous substances.
- The cell bodies are ovoid in shape and their nuclei found in a level below those of the supporting cells.
- The slender axons form bundles of nonmyelinated nerve fiber which reach the olfactory bulbs of the brain.

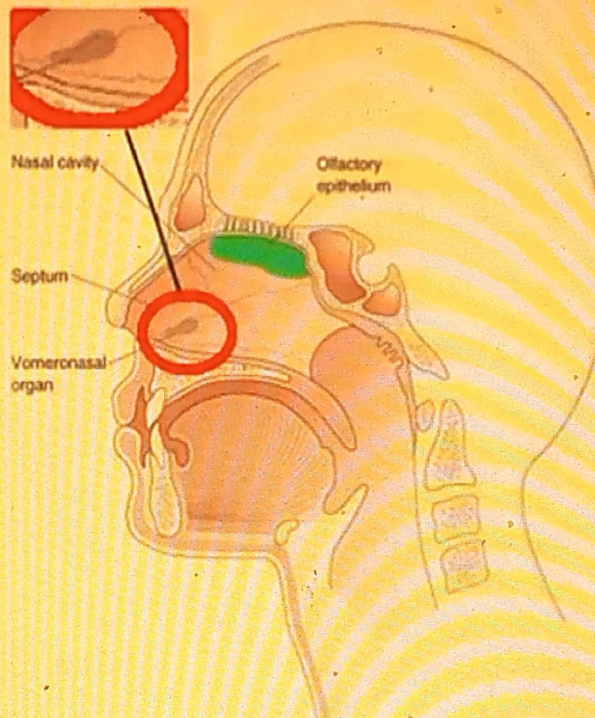
Supporting cells (sustentacular)

- They have narrow bases and wide cylindrical portions. On their free surface are microvilli submerged in a fluid layer (serous and mucous secretion) covering the entire epithelial

Vomeronasal organ

- It is a tubular blindended structure, located in the mucosa of the ventral portion of the nasal septum.

Vomeronasal Organ



-It consists of:

1. External cartilaginous support
2. Middle lamina propria
3. Internal epithelial duct

- The cartilage form J-shaped and is a hyaline type.
- The lamina propria is highly vascular and highly glandular.
- The epithelial duct (vomeronasal duct) has a crescent shape with a lateral convex and a medial concave mucosal wall. The medial epithelium is a pseudostratified columnar epithelium having basal, supporting and sensory cells.

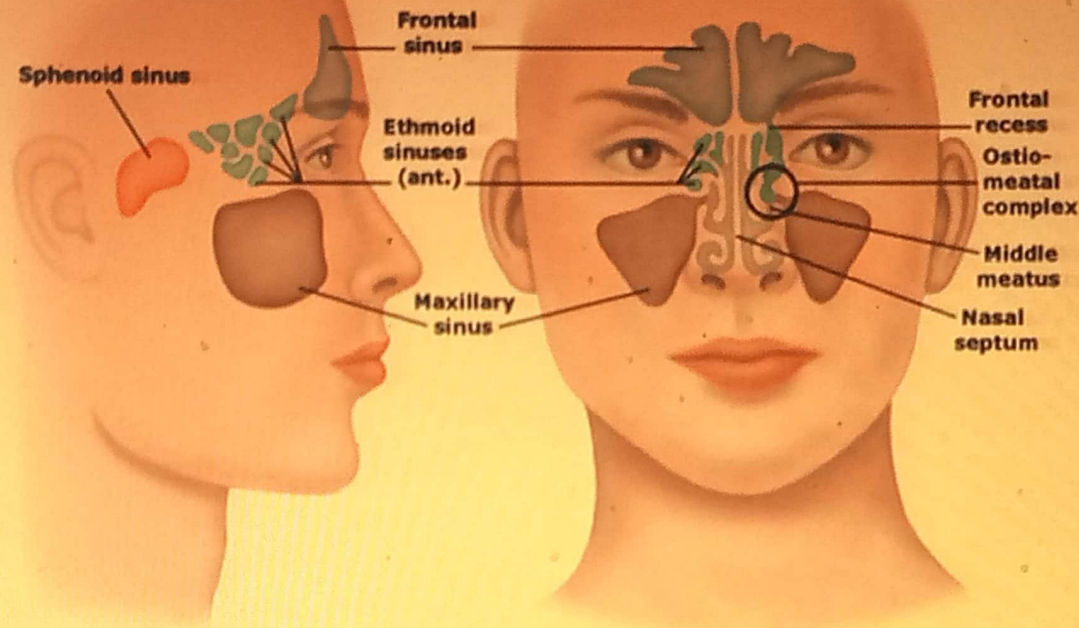
The sensory cells differ from those of the olfactory mucosa in that they lack the dendritic bulbs and have microvilli instead of cilia on their apical surfaces.

Function

- 1- It is a chemoreceptor of liquid-borne compounds of low volatility which function in sexual behavior of both male and female.
- 2- Maternal behavior of female.
- 3- Interaction of the fetus with its uterine environment.

Paranasal Sinuses

-They are blind cavities in the frontal, maxillary, ethmoid and sphenoid bones.



- The mucosa which lines these sinuses is thinner than that of the nasal cavity with which it is continuous.
- The epithelium is ciliated pseudostratified columnar with few goblet cells. The ciliary beat carries mucous toward openings connecting the sinuses with the nasal cavity.
- The lamina propria is thin layer with few glands.
- In **dog** and **cat** there are lateral nasal and maxillary recess glands in the lamina propria.

Nasopharynx

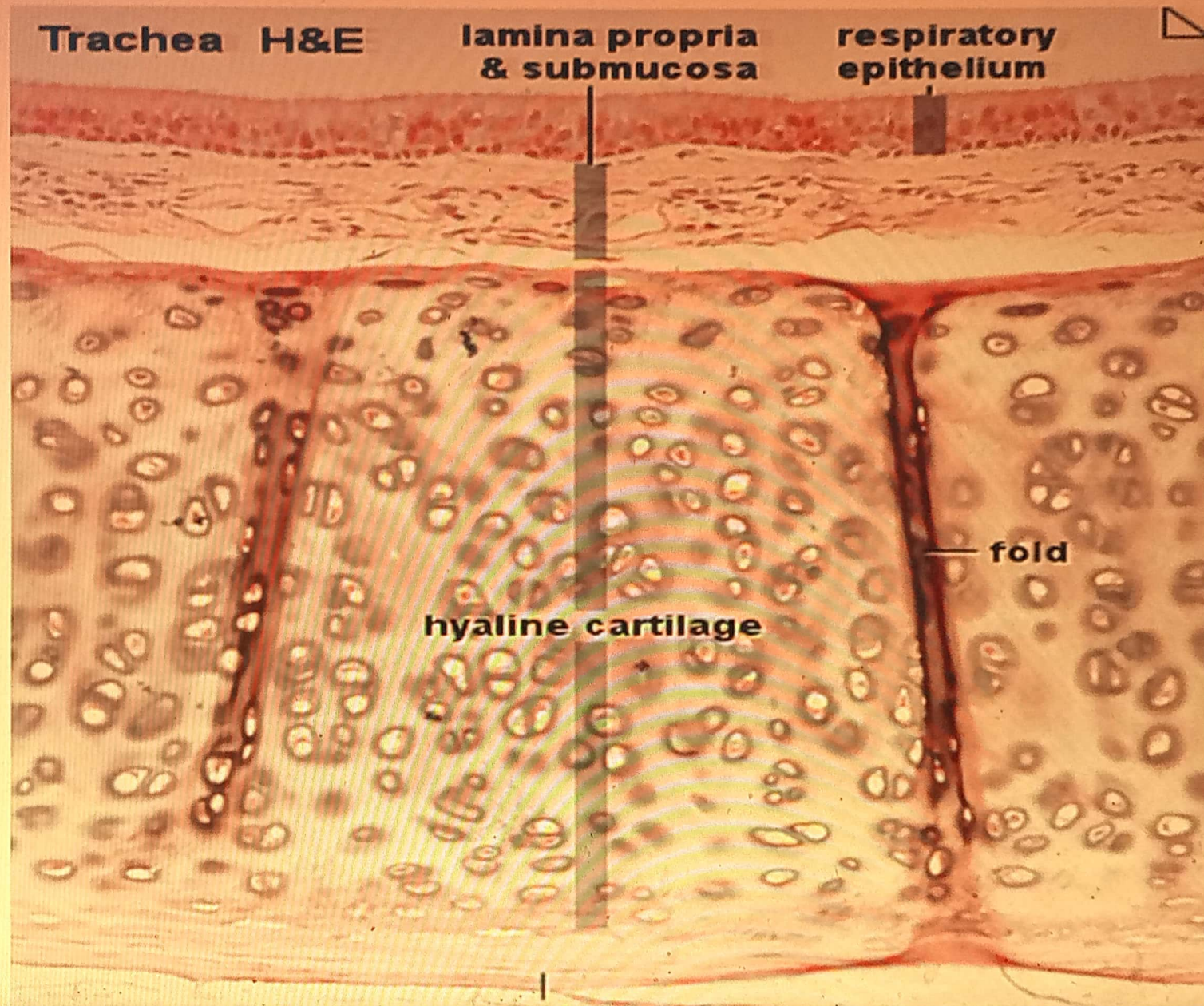
- It is that portion of the pharynx located dorsal to the soft palate and extends from the nasal cavity to the laryngopharynx.
- It is lined by ciliated pseudostratified columnar with goblet cells.

Larynx

- The larynx opens rostrally into the laryngopharynx and continuous caudally with the trachea.
- It is lined with mucosa and supported by cartilage.
- Extrinsic skeletal muscles move the larynx during swallowing and intrinsic skeletal muscles move individual cartilages during respiration and phonation.
- The mucosa of the epiglottis, laryngeal vestibule and vocal folds is lined by a non-keratinized stratified squamous epithelium. Caudal to the vocal folds the epithelium change into a ciliated pseudostratified columnar with goblet cells.
- The epithelium of the epiglottis, aryepiglottic folds and arytenoids may contain taste buds in all species except horse.

Trachea

-The trachea is a thin-walled tube that extends from the larynx to the point at which it bifurcates into two primary bronchi.



It formed of:

Tunica mucosa

- It is formed of **lamina epithelialis** and **lamina propria**.

Lamina epithelialis

The mucosa is lined by pseudostratified ciliated columnar epithelium. Seven epithelial cell types have been identified in this epithelium: basal cells, goblet cells, ciliated cells, brush cells, serous cells, clara cells and neuroendocrine cells (K cells).

Lamina propria

It is formed of loose connective tissue with dense irregular band of longitudinally arranged elastic fibers. It also contains seromucous glands, small blood and lymph vessels.

Tunica submucosa

Loose connective tissue that contains the majority of the secretory portions of the seromucous glands.

Cartilage and muscle

The **cartilage** is arranged into C- or U- shaped rings of hyaline cartilage that are incomplete dorsally. The number of these rings varies within the same species.

Tunica adventitia

It is a thin layer of dense irregular connective tissue that surrounds the cartilage rings.

Lung

- The lungs are paired organs,
- occupy most of the thoracic cavity.

Structure

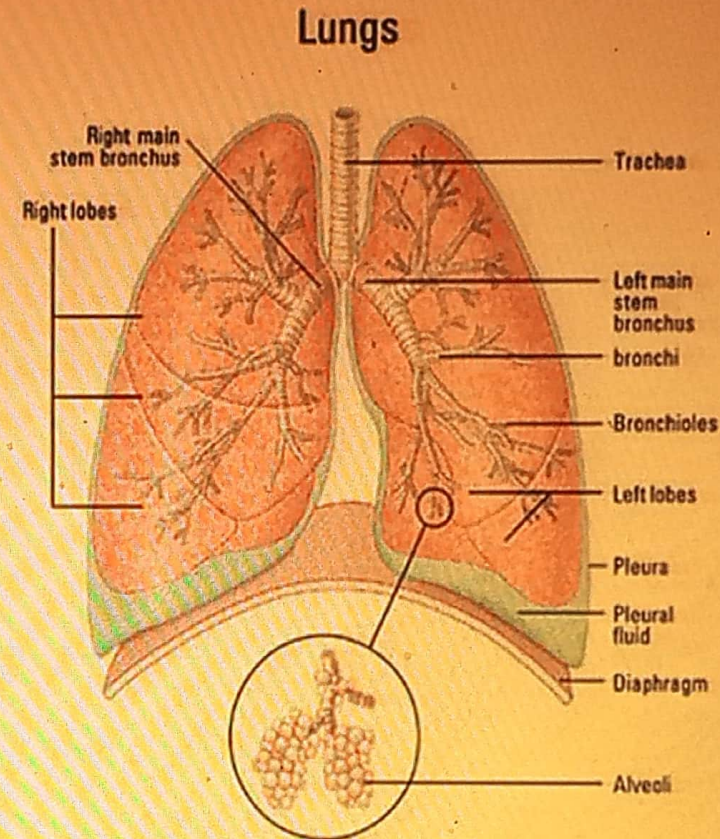
The lung consists of:

1. Pleura
2. Bronchial tree (air conducting system).
3. Respiratory portion.

Pleura

- The lung is covered by a serous membrane called **visceral pleura** (pulmonary pleura or lung capsule).

This capsule is a mesothelial cell layer rest on subserosa of dense irregular connective tissue and elastic fibers. At the periphery of the lung the connective tissue of interalveolar septa is continuous with the pleura.



At the thick areas of the pleura, the connective tissue consists of two or more layers of elastic laminae, many dense irregular collagen bundles, pulmonary capillaries and two additional sets of vessels.

-The **large domestic mammals (horse, cow)** have a very thick pleura, meanwhile **dog** and **cat** have a thin pleura.

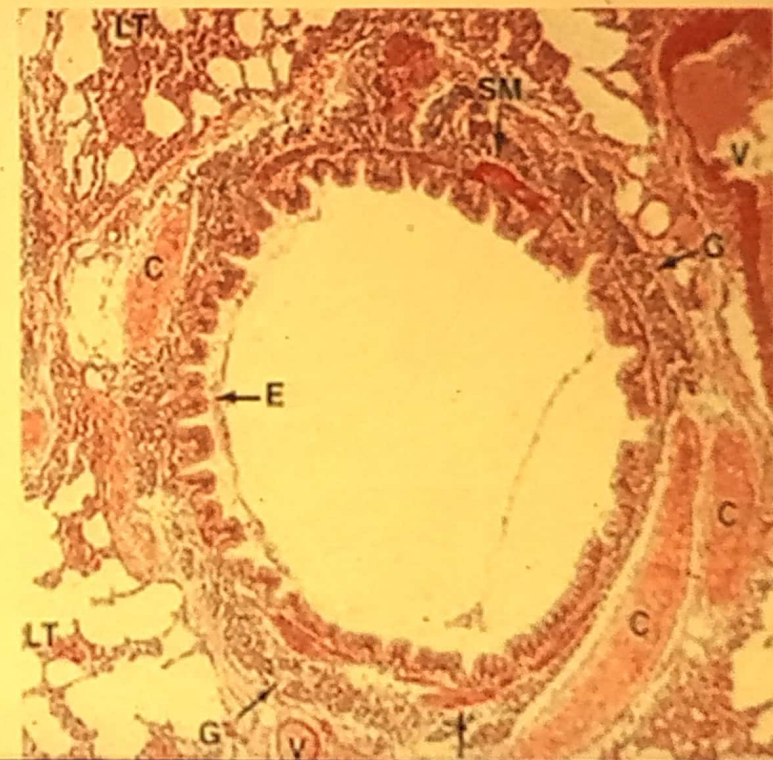
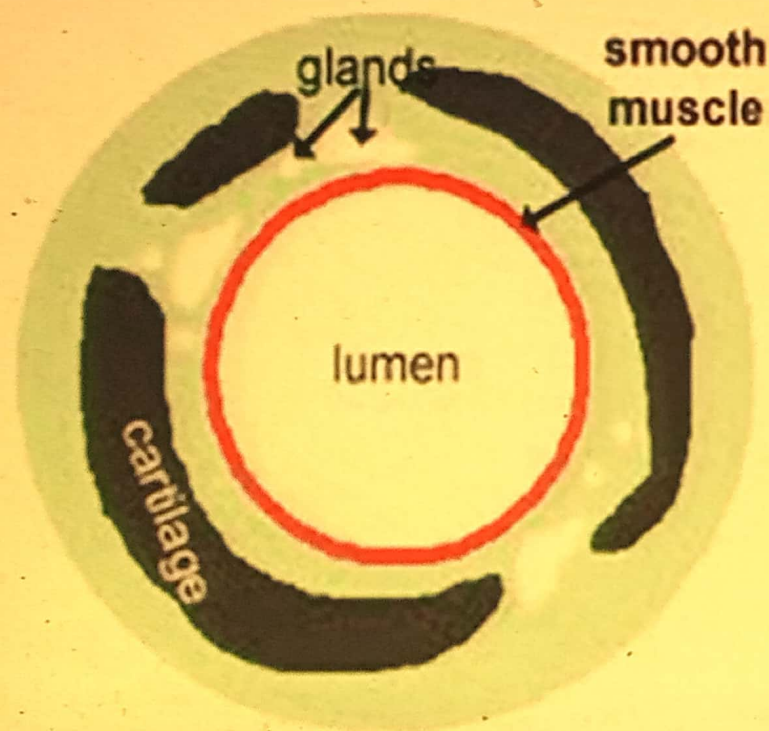
Bronchial tree

- The trachea end by dividing into two **bronchi (primary bronchi or extrapulmonary bronchi)**, each primary bronchus enter the substance of corresponding lung and divides into a number of bronchi according to the number of lobes in each lung (lobar bronchi). When the bronchus enter inside the lung it is called **intrapulmonary bronchus**. These lobar bronchi divide repeatedly, giving rise to smaller bronchi, whose terminal branches are called **bronchioles**. Each bronchiole enters a pulmonary lobule, where it branches to form five to seven **terminal bronchioles**.

-The pulmonary lobules are pyramidal-shaped with the apex directed toward the pulmonary hilum. Each lobule is delineated by a thin connective tissue septum.

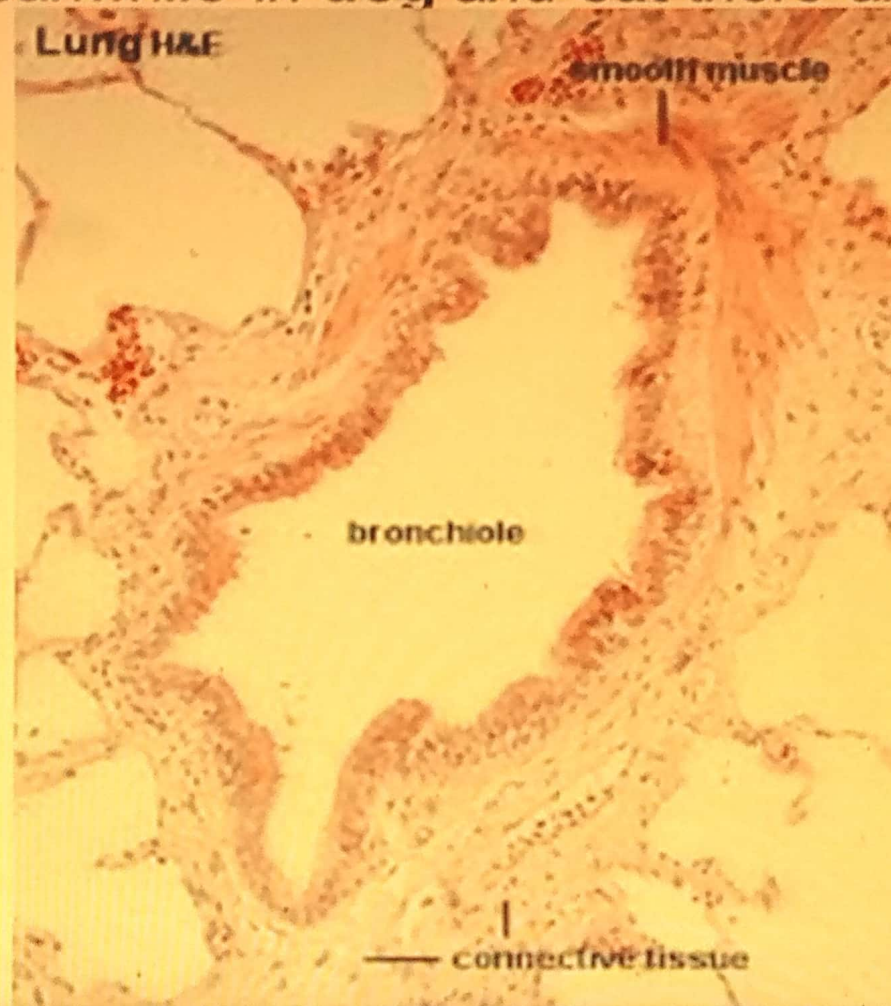
The **primary bronchi** generally have the same histological appearance as the trachea. Proceeding toward the respiratory portion, the various layers become thinner and simplified. These changes are gradual and not abrupt.

-The epithelial height and the thickness of the lamina propria decrease gradually toward the terminal branches.



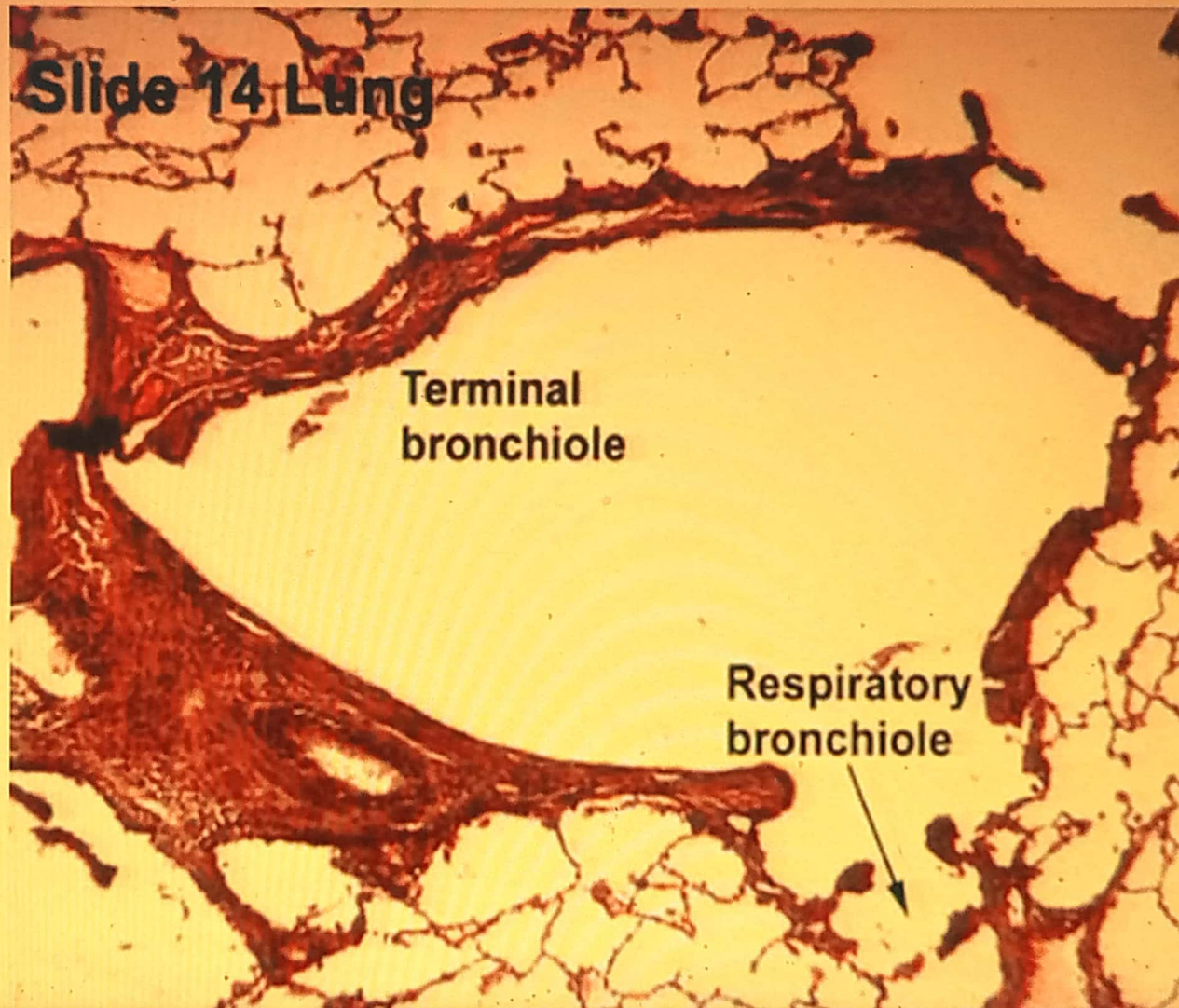
Bronchioles

- The wall of bronchiole is composed of epithelium, smooth muscle and small amounts of connective tissue **without glands or cartilage**.
- In **horse, ox** and **sheep** there are several generations of bronchioles meanwhile in **dog** and **cat** there are only one or two generations.



Terminal bronchiole

- Lamina, epithelialis the epithelium changes from pseudostratified ciliated columnar in the large bronchiole into simple columnar or cuboidal in the terminal bronchioles. This epithelium is composed of ciliated cells and clara cells.



Clara cells (bronchiolar exocrine cells)

These cells have characteristics of both secretory cells and cell capable of metabolizing xenobiotic compound.

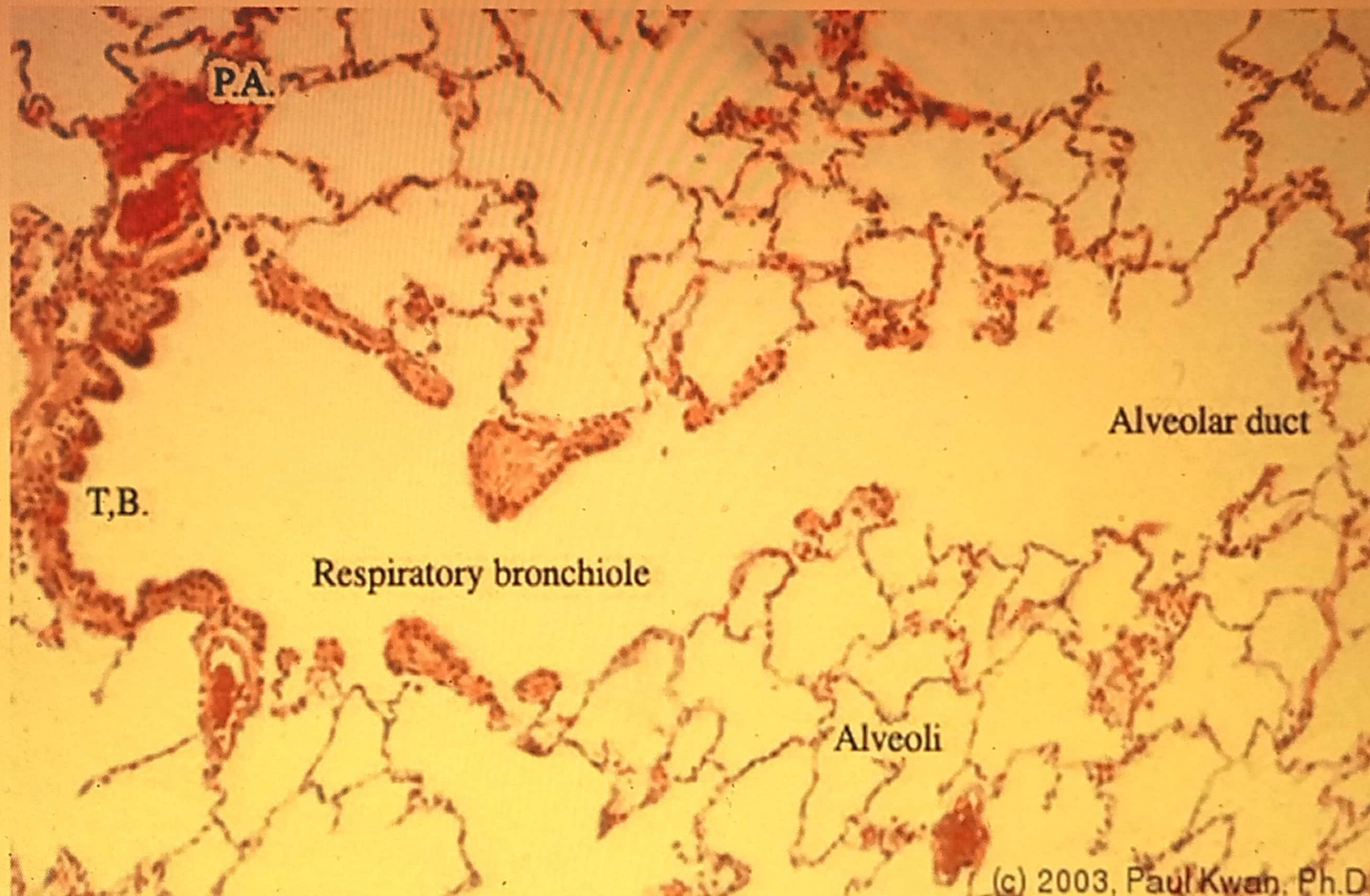
Lamina propria and submucosa is extremely reduced and **Muscle**, separate fascicles that branch and anastomose circularly and obliquely.

Adventitia, thin layer of loose connective tissue.

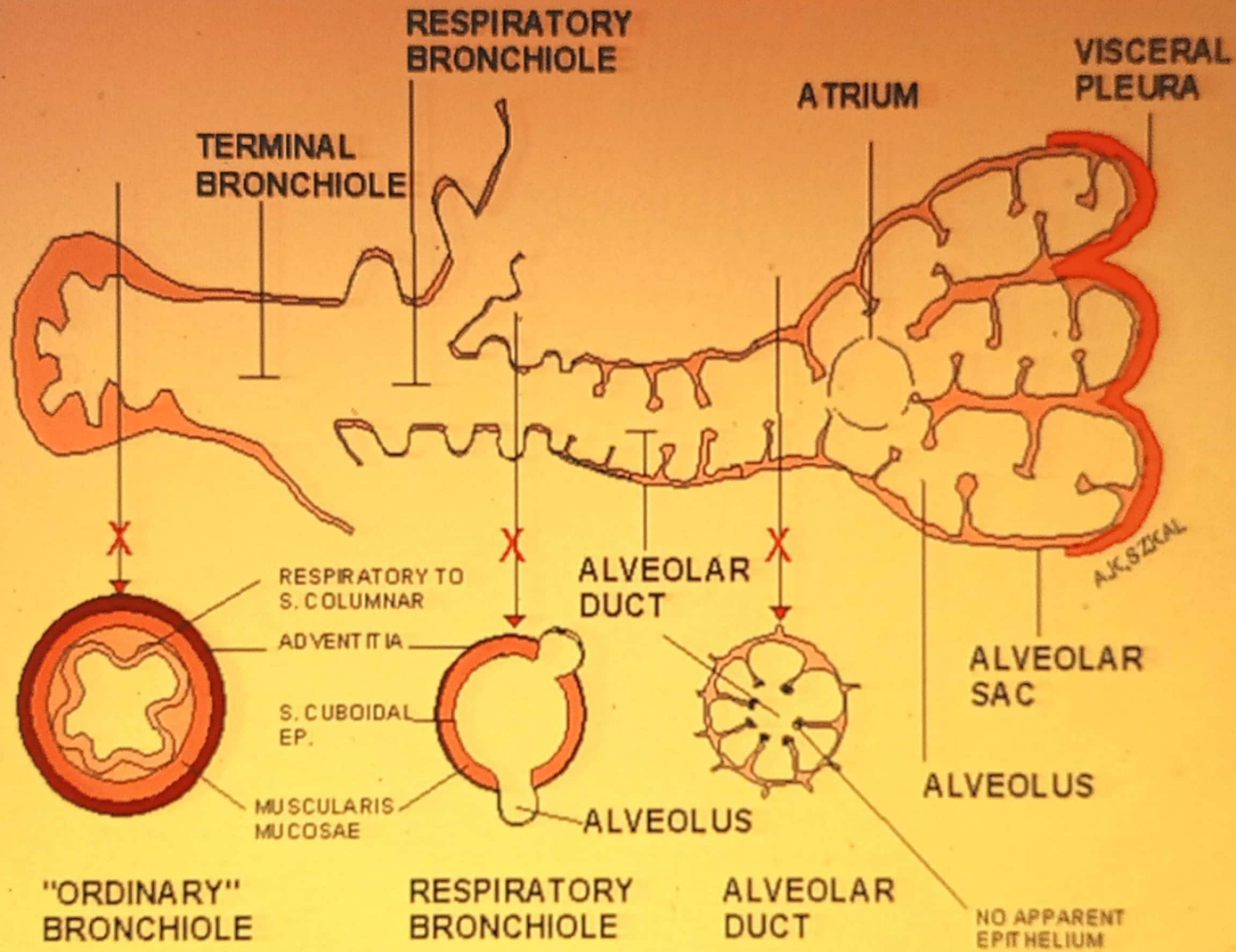
formed of loose connective tissue.

Respiratory Portion

- This is the area where gas exchange takes place and consists of **respiratory bronchioles**, **alveolar ducts**, **alveolar sacs** and **alveoli**.



HISTOLOGY OF BRONCHIOLES



- The functional unit of the gas exchange area is called **acinus**. It includes respiratory bronchioles, alveolar ducts, alveolar sacs and alveoli.
- In many species, acini are separated into groups, called **lobules**.
- The lobule is a structural unit and comprises a cluster of acini that is separated from adjacent clusters by connective tissue **septa** (interlobular septa) formed of collagen and elastic fibers and blood vessels.

Respiratory bronchiole

- It is very similar in structure to the terminal bronchiole except that the epithelium is interrupted by alveoli.
- The smooth muscle cells underlie the simple cuboidal or columnar and the alveoli open between these muscle cells.
- In **horse, cow, sheep, pig** and some **rodents** the respiratory bronchiole is very short or absent, but in **dog** and **cat** it is clear with extensive alveolarization and its epithelium consists almost entirely of clara cells.

Alveolar ducts and alveolar sacs

- One to five generations of alveolar ducts are supplied by a single respiratory bronchiole
- The alveolar duct terminates in cluster of alveoli called **alveolar sac**.

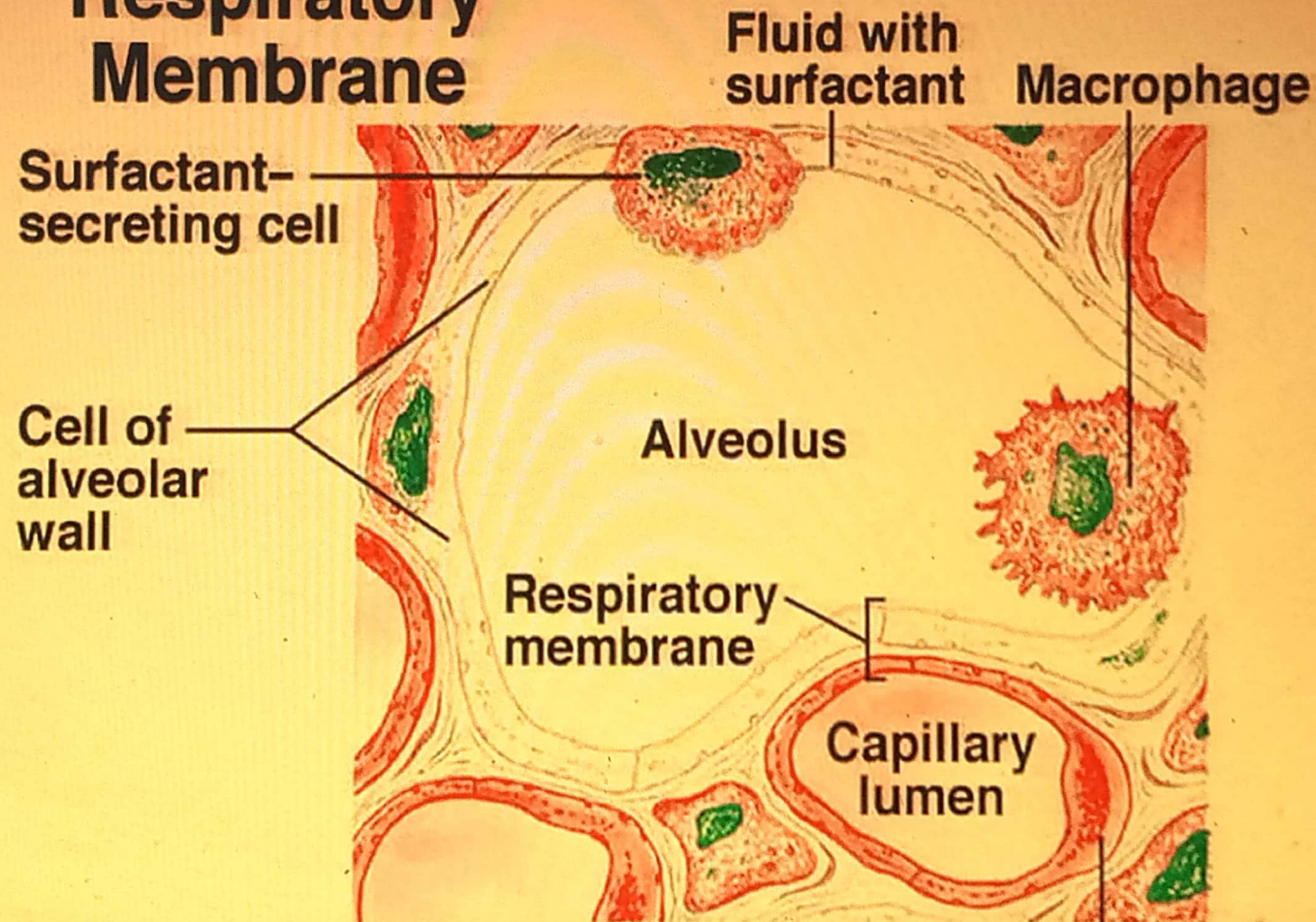
Atrium, is a shared space into which a number of alveolar sacs open.

Alveoli

- The alveoli are saclike airspaces that open into alveolar sac, alveolar duct or respiratory bronchiole and are separated by interalveolar septa
- The alveoli are lined by two types of epithelial cells:

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Respiratory Membrane



1- Squamous or type I cell

- It is very flat cell with flattened centrally located nucleus and few organelles in its cytoplasm.
- This cell type cover 97% of the alveolar surface area and tight junctions and desmosomes are found between neighboring cells.
- These cells rest on a continuous basal lamina and serve as a very thin gas permeable component of the blood air barrier.

2- Type II or great alveolar cell

- It is cuboidal cell with central nucleus that covers 3% of the alveolar surface area.
- It is filled with large numbers of organelles and its free surface has large numbers of microvilli.

It is characterized by the presence of large numbers of **osmiophilic lamellar bodies**. These lamellar bodies are the source of the pulmonary surfactant that lines the alveoli as a thin film to reduce the surface tension and prevent them from collapsing. It also has a bactericidal effect.

- Type II cells also function as the progenitor cells for both type I and type II cells.

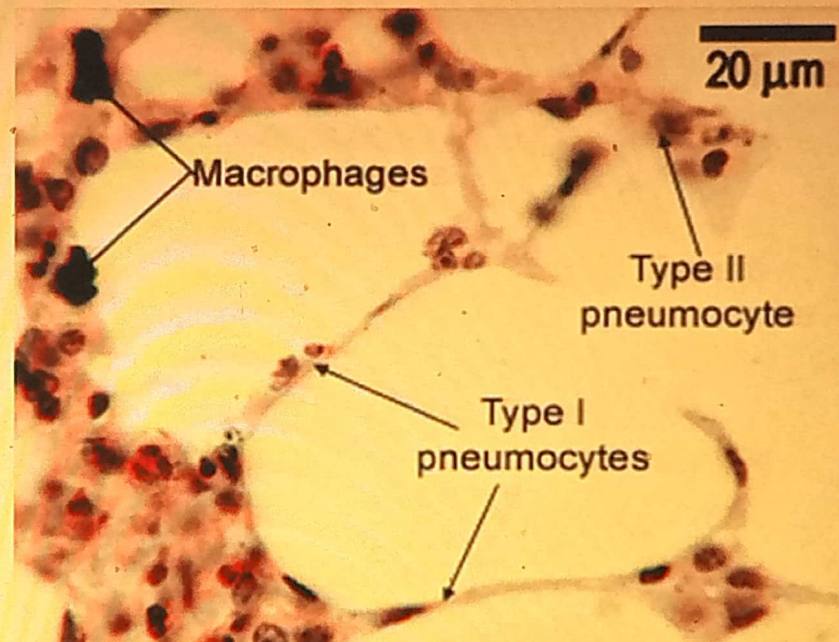
Blood air barrier

- It is that barrier through which gas exchange takes place between the air in the lumen of the alveoli and blood in capillaries. It is about $1.5\text{ }\mu\text{m}$ thick and formed of:

- 1- Film of pulmonary surfactant on alveolar surface .
- 2- Cytoplasm of type I alveolar cell.
- 3- Fused basal laminae of type I cell and capillary endothelium.
- 4- The cytoplasm of capillary endothelium (not fenestrated).

Alveolar macrophages

- They are large phagocytic cells found on the air side of the alveolar surface. They are considered one of the mononuclear phagocytic system (develop from monocytes).
- When they phagocytose small dust particles they are called **dust cells**. When they phagocytose red blood cells they are called **heart failure cells**.



Alveolar pores, are openings in the interalveolar septa interconnect adjacent alveoli. They are lined by epithelial cells and permit air and alveolar macrophages

Pulmonary circulation

2- Systemic circulation

- Bronchial arteries (from the aorta) carry oxygenated blood and nutrients to the lung tissues.
- Bronchial veins carry unoxygenated blood.

Lymphatic drainage

- The lung's lymphatic vessels are divided into superficial and deep networks. Both drain toward lymph nodes near the hilum.
- Lymph vessels are absent from interalveolar septa.

Innervation

- Parasympathetic motor fibers (vagus) stimulate bronchial constriction.
- Sympathetic fibers cause bronchial dilatation.

Thank you